Tutorial #2: Statistical Regression

1. Suppose we have data $X_1, X_2, \ldots, X_n$ that are i.i.d as $X_i \sim \text{Binomial}(m, p)$, where m is known and p is the unknown success probability.

   (a) Which prior distribution would you choose for p? Justify your choice.

   (b) Derive the posterior distribution of p given the data.

   (c) If your prior is $p \sim \text{Beta}(2, 5)$ and you observe data where in $n = 10$ experiments, the total number of successes is $\sum_{i=1}^{10} X_i = 25$ (with $m = 5$ trials per experiment), what is the posterior distribution?

   (d) Compute the posterior mean and posterior variance.

2. Consider a model with two parameters θ_1 and θ_2 with the following full conditional distributions:

$$\theta_1|\theta_2, \mathbf{X} \sim \text{Gamma}(3, 2 + \theta_2)$$

$$\theta_2|\theta_1, \mathbf{X} \sim \mathcal{N}(\theta_1, 1)$$

   (a) Describe the steps of the Gibbs sampling algorithm for this model.

   (b) Write out the first three iterations of Gibbs sampling starting from initial values $\theta_1^{(0)} = 1$, $\theta_2^{(0)} = 0$.

   (c) What diagnostic tools would you use to check if your Gibbs sampler has converged?

3. Consider the simple linear regression model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma^2), \quad i = 1, \ldots, n$$

   (a) Write the likelihood function $L(\beta_0, \beta_1, \sigma^2|\mathbf{Y}, \mathbf{X})$.

   (b) Derive the log-likelihood function.

   (c) What prior distributions would you choose for β_0, β_1, and σ^2 in a Bayesian analysis?

   (d) Would you use the likelihood or log-likelihood when computing the posterior distribution? Explain why.

4. You run an MCMC algorithm and obtain the following trace plots for two different parameters:

   Parameter A: The trace plot shows the chain quickly moving to a region and oscillating randomly around a stable value.
   Parameter B: The trace plot shows the chain slowly drifting upward over many iterations without stabilizing.

   (a) For each parameter, comment on whether the chain appears to have converged.

   (b) What steps would you take to improve convergence for Parameter B?

   (c) How would you determine an appropriate burn-in period for each chain?

5. Suppose you have a posterior sample of 10,000 draws for a regression coefficient β after burn-in.

   (a) How would you construct a 95% credible interval for β?

   (b) If the 95% credible interval is $(-0.15, 0.08)$, what can you conclude about the significance of β?

   (c) What is the difference between a credible interval and a confidence interval?

   (d) How would you compute the posterior probability that $\beta > 0$?

6. Conceptual Questions

   (a) What is the most commonly used estimator in Bayesian statistics, and how is it approximated using MCMC samples?

   (b) Why is it important to use a burn-in period in MCMC, and how does it help with convergence?

   (c) Explain what conjugate priors are and provide two examples of conjugate prior-likelihood pairs.

   (d) In Bayesian regression, how do we determine if a coefficient is significantly different from zero?

   (e) If you obtain a p-value of 0.008 in regression, what can you conclude about the coefficient's significance?

   (f) What is the relationship between credible intervals and hypothesis testing in Bayesian analysis?